Detecting Hypoglycemia from ECG Signals Using Deep Learning

Background/Problem

Hypoglycemia happens when a person's blood sugar drops below 70 mg/dL. It can cause dizziness, confusion, seizures, or even loss of consciousness if not detected early. People with diabetes, especially those using insulin, are at high risk. Current monitoring methods like fingerstick tests and Continuous Glucose Monitoring (CGM) are invasive and uncomfortable for daily use. Research has shown that low blood sugar can cause small changes in heart activity, which can be seen in ECG signals. This makes it possible to detect hypoglycemia using ECGs in a non-invasive way.

Objectives

1. Build a deep learning model that can detect when a patient is hypoglycemic based on 10-second ECG recordings of 2500 samples.
2. Test if ECG signals alone can give early warning of low blood sugar.
3. Make sure the model works well for different patients and avoids data leakage.

Brief Method

The plan is to ensure the data from the ECG signal, which is 2500 samples over a 10 second period for 9 different patients, is clean and normalized. We plan to clean the data by removing the following data points: patient ID, count, day and glucose_values as they are all irrelevant to the ECG signal data. We need only the samples and the true labels whether they experienced hypoglycemia or not in order to train our neural network. We plan to use either BiLSTM, or hybrid 1d CNN-BiLSTM model. The 1D CNN layers would pick-up short-term ECG patterns, then the BiLSTM layers will see how the signal changes over time. We will split the data by patients, 7 for training, 1 for validation, and 1 for testing to prevent overlaps between patients. This is subject to change as per the results during training. We also plan to do multiple rounds, each round having a different patient for validation or for testing, implementing a cross validation to ensure robustness. Model performance will be measured using accuracy, sensitivity, specificity, F1-score. We will also apply dropout and early stopping to prevent overfitting.

Expected Outcome

A reliable deep learning model that can detect hypoglycemia from ECG signals, showing that ECG can be used as a non-invasive way to monitor glucose levels.